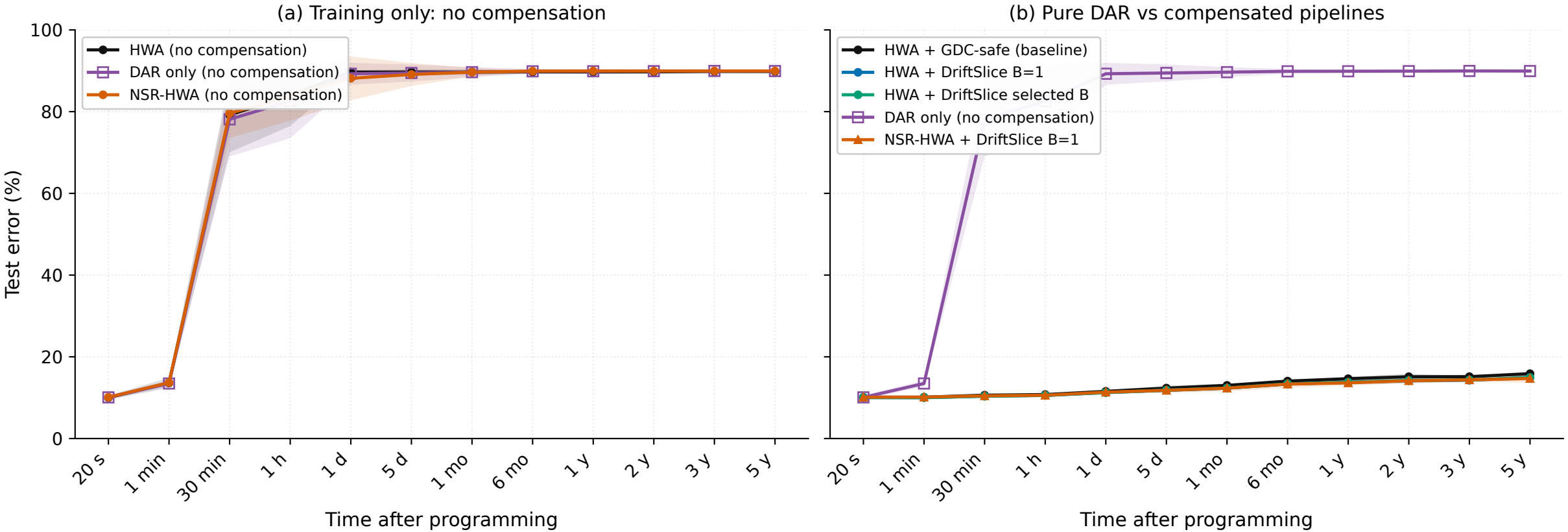
ResNet-32 / CIFAR-10: test error under PCM drift
(a) Training only: no compensation
(b) Pure DAR vs compensated pipelines
Test error (%)
100
80
60
40
20
0
HWA (no compensation)
DAR only (no compensation)
NSR-HWA (no compensation)
HWA + GDC-safe (baseline)
HWA + DriftSlice B=1
HWA + DriftSlice selected B
DAR only (no compensation)
NSR-HWA + DriftSlice B=1
20 s
1 min
30 min
1 h
1 d
5 d
1 mo
6 mo
1 y
2 y
3 y
5 y
Time after programming
Time after programming
Mean ± 2 SD across 3 training repetitions; 30 device realizations per repetition
Timestamps are equally spaced · bands describe variability, not 95% confidence intervals